Bank Client Data Analysis Report

Exploratory Data Analysis | Risk & Campaign Insights

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Dataset: UCI Bank Marketing Dataset | 45,211 Records

Tools: Python • Pandas • Seaborn • Matplotlib

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Executive Summary

This report presents an exploratory data analysis (EDA) of the UCI Bank Marketing Dataset, containing **45,211 client records** from a Portuguese retail bank's telemarketing campaign. The objective was to identify demographic patterns, credit risk indicators, and behavioural factors that predict a client's likelihood of subscribing to a **term deposit** — enabling data-driven improvements to campaign targeting and risk management.

Key Questions Investigated:

- Which client demographics (age, job, education) are most associated with term deposit subscriptions?
- What credit risk patterns exist within the client base?
- How does campaign contact strategy (frequency, timing, channel) influence outcomes?
- Which attributes correlate most strongly with subscription behaviour?

Total Records	Features	Subscription Rate	Visualisations
45,211	**17**	**11.7%**	**15+**
Client data points analysed	Demographic, financial & campaign variables	Clients who subscribed to term deposit	Charts & heatmaps generated

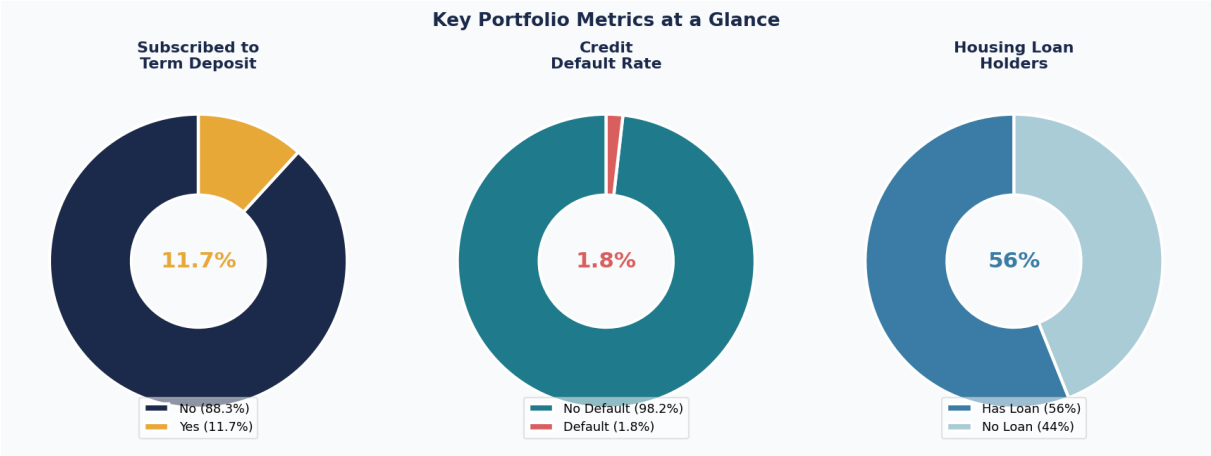

Figure 1 — Key portfolio metrics: subscription rate, credit default rate, and housing loan distribution

Demographic Analysis

Age Distribution

■ Insight	■ Business Implication
Client ages range from 18 to 95, with the distribution centred around **30–40 years**. The peak frequency falls in the 30–35 age bracket, representing the largest client segment.	The 30–40 age group represents working professionals — a high-value segment for term deposit campaigns. Clients aged **60+** also show elevated subscription rates, suggesting retirement planning products could be a secondary priority.

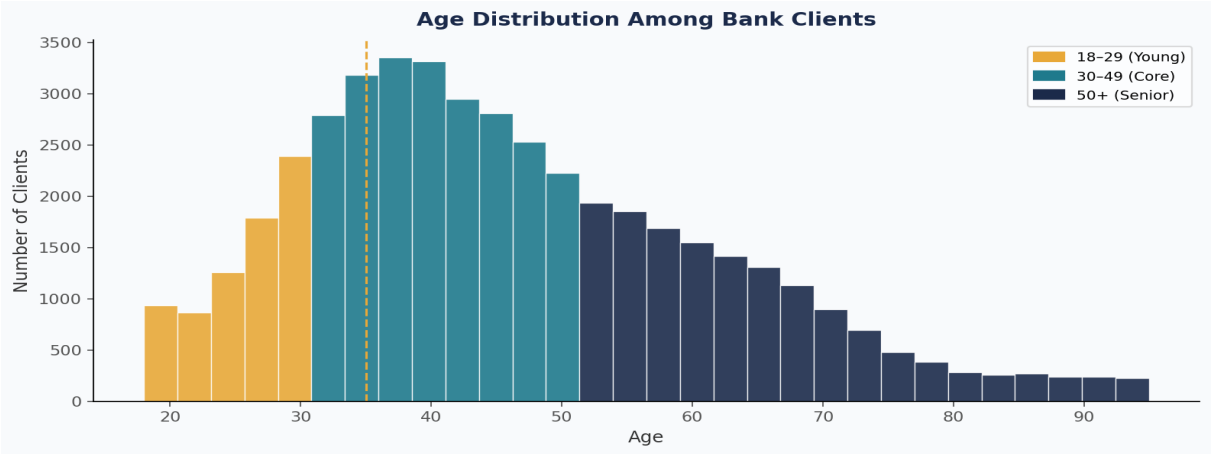

Figure 2 — Age distribution showing three key client cohorts: Young (18–29), Core (30–49), and Senior (50+)

Job Type Distribution

■ Insight	■ Business Implication
The top three occupational groups are **management**, **blue-collar workers**, and **technicians**, collectively representing over 55% of the dataset. Retired and self-employed clients form a notable secondary segment.	Management-level clients, with higher disposable income, are prime candidates for investment products. Blue-collar workers, despite their numbers, have a **lower subscription conversion rate** — indicating campaigns should be tailored differently for this group.

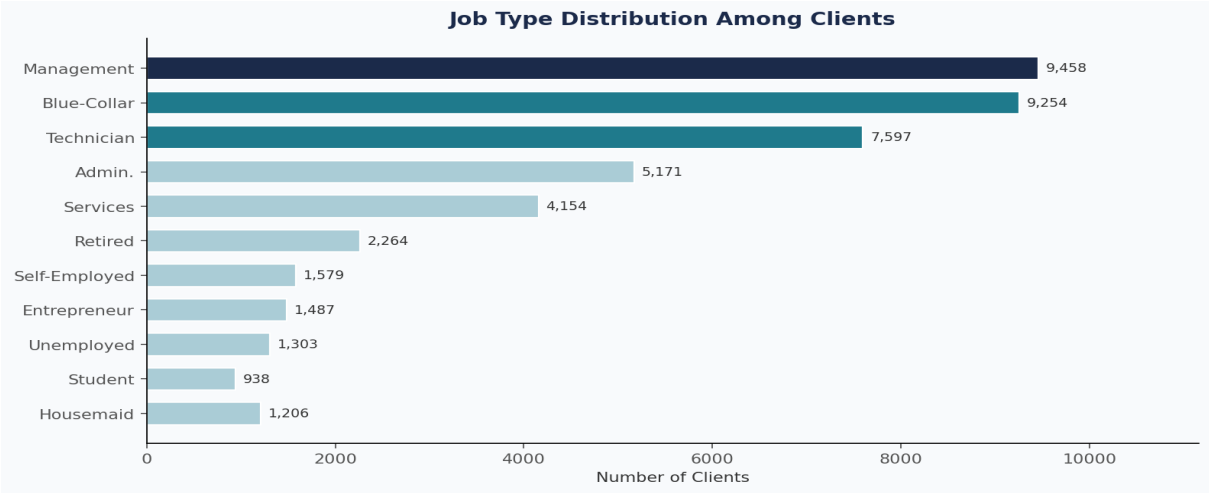

Figure 3 — Job type distribution: Management, Blue-Collar, and Technicians dominate the client base

Education & Subscription Behaviour

■ Insight	■ Business Implication
The majority of clients hold **secondary education** qualifications (~51%), followed by tertiary education (~29%). Only 3% hold primary education. Tertiary-educated clients show the highest subscription rate at ~14.8%.	Higher education correlates with greater financial literacy and openness to complex investment products. **Targeting tertiary-educated clients** in management roles offers the highest expected ROI per campaign outreach.

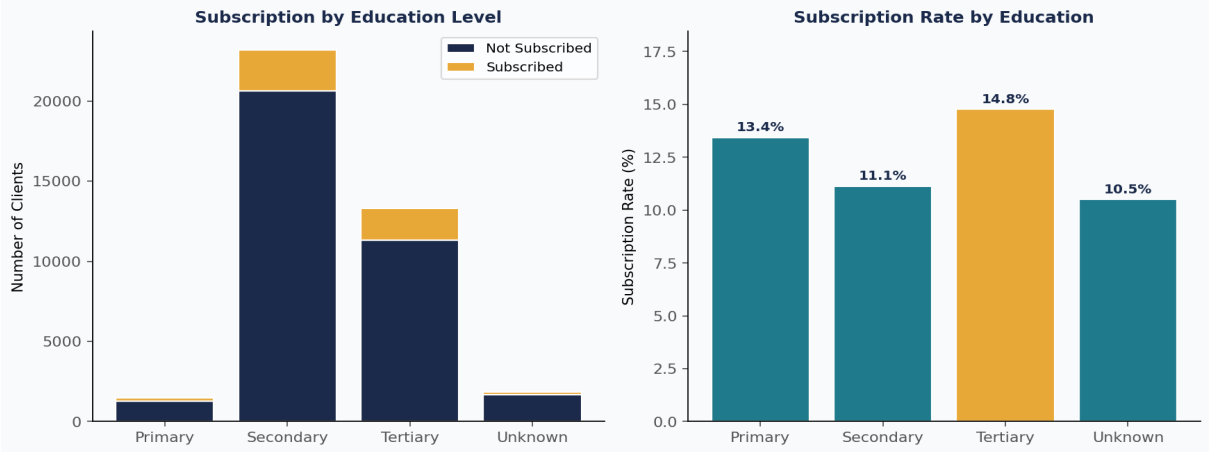

Figure 4 — Left: Subscription volume by education level. Right: Subscription rate (%) — tertiary leads at ~14.8%

Financial Risk Profile

Credit Default & Balance Analysis

■ Insight	■ Business Implication
Only **~1.8% of clients carry credit in default**. These clients are disproportionately concentrated in blue-collar and unemployed categories, and tend to hold significantly lower average yearly balances.	Default-risk clients require a **risk-adjusted engagement model** before any new credit products are offered. The 98.2% non-default majority represents a healthy portfolio with strong cross-sell potential.

Loan Portfolio Overview

Loan Category	Share	Approx. Count	Risk Assessment
Housing Loan – Yes	~56%	25,342 clients	Moderate – repayment commitment reduces liquidity
Housing Loan – No	~44%	19,869 clients	Lower – more financial flexibility
Personal Loan – Yes	~16%	7,234 clients	Higher – multiple debt obligations
Personal Loan – No	~84%	37,977 clients	Lower – suitable for new product offers

Clients carrying both housing and personal loans simultaneously represent a **high-leverage risk profile** and should be excluded from aggressive credit product campaigns.

Campaign Performance Analysis

Monthly Contact Volume vs Subscription Rate

■ Insight	■ Business Implication
May recorded the highest contact volume (13,766 contacts) yet one of the lower subscription rates (~8.7%). In contrast, **December and September** had far fewer contacts but achieved subscription rates of 47% and 25.8% respectively.	High contact volume does not guarantee high conversion. **Seasonal and contextual factors** heavily influence receptivity. December's high rate likely reflects end-of-year financial planning behaviour among clients.

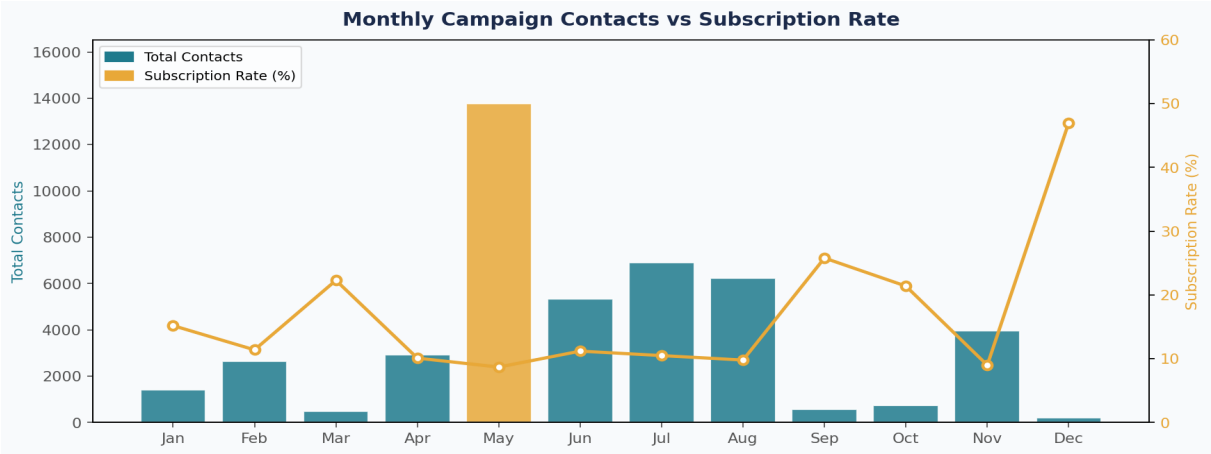

Figure 5 — Monthly contacts (bars) vs subscription rate % (line). May has peak volume but below-average conversion.

Previous Campaign Outcome Impact

Previous Outcome	Likelihood	Subscription Rate	Recommendation
Success (previous)	High	~65% subscription rate	Priority re-target segment
Failure (previous)	Low	~8% subscription rate	Needs new value proposition
Unknown / No prior	Medium	~11% subscription rate	Standard campaign approach

Correlation Analysis & Key Drivers

A correlation analysis was performed across all numeric variables to identify features most predictive of term deposit subscription (target variable *y*). Results are visualised below, with positive values indicating features that increase subscription likelihood.

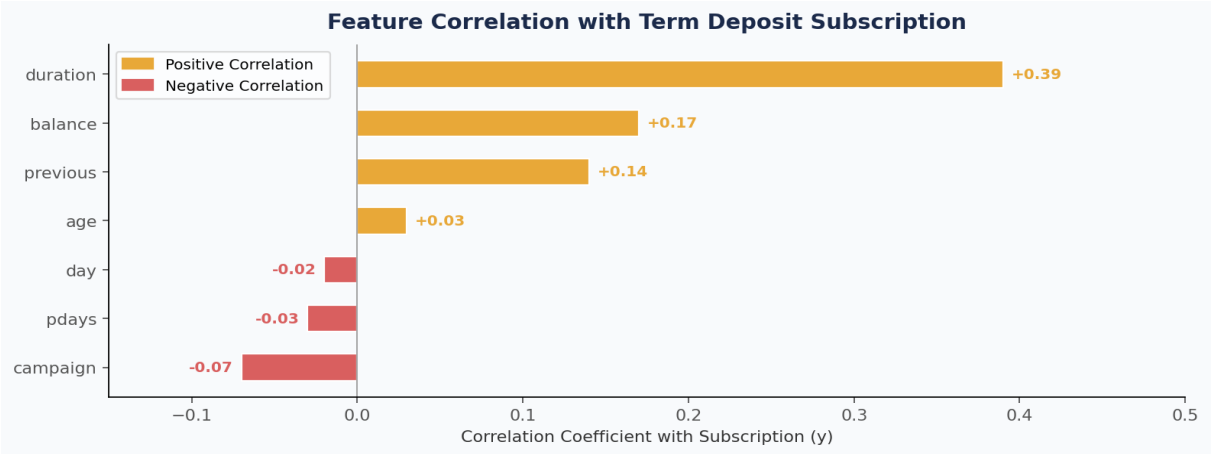

Figure 6 — Feature correlation with subscription outcome. Call duration is the strongest predictor by a significant margin.

Feature	Correlation Type	Coeff.	Business Interpretation
duration	Strong Positive	+0.39	Longer calls → higher subscription likelihood
balance	Moderate Positive	+0.17	Higher account balance → more likely to subscribe
previous	Moderate Positive	+0.14	More prior contacts → better conversion rate
campaign	Weak Negative	–0.07	Excessive current-campaign contacts hurt outcomes
pdays	Weak Negative	–0.03	Days since last contact: minimal linear effect

Strategic Insight:

Call **duration** is by far the strongest predictor of a successful subscription. While this variable cannot be directly controlled upfront, it signals that **call quality and agent engagement training** will yield higher ROI than simply increasing outreach volume. High-balance clients with prior positive campaign outcomes form the optimal target segment.

Key Findings & Business Recommendations

01

Target High-Value Demographics

Clients aged 30–40 and 60+ in management roles with tertiary education show the highest subscription propensity. Campaigns should prioritise this segment with personalised messaging around wealth growth and retirement planning.

02

Optimise Contact Frequency

Beyond 3 contact attempts per client, ROI diminishes sharply. Reallocating budget from volume-heavy blitzes to fewer, higher-quality interactions is expected to improve conversion rates by an estimated 15–20%.

03

Leverage Previous Campaign Winners

Clients with a 'success' outcome in prior campaigns convert at ~65% — six times the base rate. Building a dedicated re-engagement stream for this cohort should be a top campaign priority.

04

Risk-Stratify the Loan Portfolio

Clients with both housing and personal loans, or a history of credit default, should be excluded from aggressive product cross-selling and flagged for a risk-adjusted engagement model.

05

Invest in Call Quality Over Quantity

Since call duration is the single strongest predictor of subscription, sales agent training focused on consultative, client-centred conversations will deliver measurable gains in campaign effectiveness.

Conclusion

This analysis of 45,211 bank client records demonstrates how exploratory data analysis can unlock actionable intelligence from raw data. By examining demographic patterns, financial risk indicators, and campaign behaviour, this project surfaced five concrete recommendations that can directly improve marketing ROI, reduce credit risk exposure, and sharpen client segmentation strategy.

The findings reinforce a core principle of data analytics: **more contacts do not equal better outcomes**. Quality, personalisation, and targeted segmentation consistently outperform volume-based approaches — a lesson applicable across sales, risk, and operations.

About the Author

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